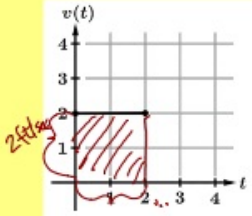
CHA-4.A.1 The area of the region between the graph of a rate of change function and the x axis gives the accumulation of change.
CHA-4.A.2 In some cases, accumulation of change can be evaluated by using geometry.
CHA-4.A.3 If a rate of change is positive (negative) over an interval, then the accumulated change is positive (negative).
CHA-4.A.4 The unit for the area of a region defined by rate of change is the unit for the rate of change multiplied by the unit for the independent variable.

6-1 Accumulations of Rates

Example 1: A particle P moves along a straight path with a constant velocity of 2 ft/sec. The graph of $v_P(t)$ is given at the right. What is the total distance the particle traveled from $t = 0$ to $t = 2$ seconds?

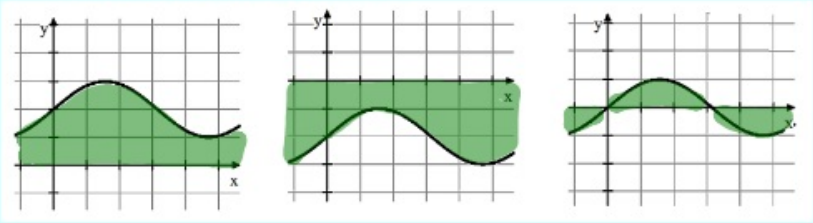
$$\text{distance} = \left(2\,\frac{\text{ft}}{\text{sec}}\right) \times (2\,\text{sec})$$
$$= 4\,\text{ft}$$



Graph of $v_P(t)$

Area Under the Curve	The region between a function and the $x-$ axis is called the **<u>area under the curve</u>**. "Under" does **<u>not</u>** mean below, it means between the function and the $x-$ axis.

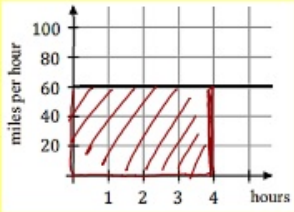
Shade the area under the curves:



Accumulation of Change	The area bounded by a **rate** function gives the **accumulation of change**. This accumulated change can be positive (above the $x-$ axis) or negative (below the $x-$ axis).

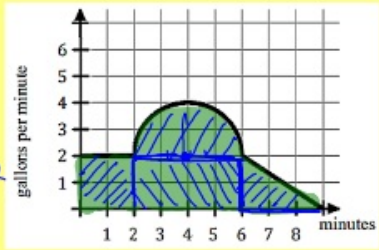
Example 2: You are on a road trip and have your car on cruise control for 4 hours. You travel at 60 miles per hour. How far have you traveled?

$$\text{distance} = \left(60\,\frac{\text{mi}}{\text{hr}}\right)(4\text{ hr})$$
$$= 240\text{ mi}$$



UNITS for Area Under the Curve	The units can be found by using multiplication: the dependent units $\times$ the independent units or the units for y $\times$ the units for x

Example 3: The graph at the right represents the rate at which water is leaking out of a tank. The units are given in the graph. How much water has leaked out of the tank after 9 minutes?



$$\text{Units} = \frac{\text{gal}}{\text{min}} \times \text{min}$$
$$= \text{gallons}$$

$$\text{Amt of Water} = 2(2) + 4(2) + \tfrac{1}{2}\pi(2)^2 + \tfrac{1}{2}(3)(2)$$
$$= 4 + 8 + 2\pi + 3$$
$$= \underline{15 + 2\pi \text{ gallons}}$$

Example 4: A particle is moving along the $x-$ axis at a rate modeled by $r(t)$ and shown in the graph at the right. The particle is at the origin at $t = 0$.

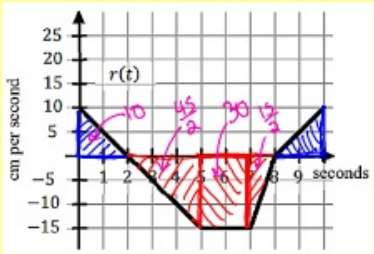
(a) How far is the particle from its starting position after 10 seconds?

displacement

$= \frac{1}{2}(2)(10) - \frac{1}{2}(3)(15) - 2(15)$
$- \frac{1}{2}(1)(15) + \frac{1}{2}(2)(10)$

$= 10 - \frac{45}{2} - 30 - \frac{15}{2} + 10$

$= -10 - 30 = -40$ cm 40cm to the left



(b) How far is the particle from its position at $t = 2$ after $t = 8$ seconds? What is the displacement of the particle from $t=2$ to $t=8$ seconds?

Position at $t=2$: $\frac{1}{2}(2)(10) = 10$cm

Position at $t=8$: $10 - \frac{45}{2} - 30 - \frac{15}{2} = 10 - 30 - 30 = -50$cm

Position at $t=8$ − Position at $t=2$ $= -50 - 10 = -60$cm

60cm to the left